

VMAF-Aware Runtime Shielding for Constrained Deep RL in Adaptive Video Streaming

Saeed Zarbi^a, Leili Farzinvash^a, Pedram Salehpour^{a,*}

^a*Department of Computer Engineering, Faculty of Electrical and Computer Engineering,
University of Tabriz, Tabriz, Iran*

Abstract

Deep Reinforcement Learning (DRL) policies for client-side Adaptive Bitrate (ABR) streaming can maximize average Quality of Experience (QoE), yet they concentrate catastrophic rebuffering risk in the tail—particularly under 5G/mmWave throughput volatility. Conventional runtime shields mitigate stalls by uniformly decrementing the bitrate index, but this content-blind strategy sacrifices perceptual fidelity disproportionately because adjacent rungs on the encoding ladder can differ by an order of magnitude in VMAF.

We formulate ABR as a Constrained Markov Decision Process (CMDP) and solve it with Lagrangian PPO augmented by manifest lookahead and buffer-stabilization cues. On top of the learned policy we introduce a *VMAF-aware runtime shield*: a deterministic projection layer that, upon emergency step-down, searches the bitrate ladder under buffer-feasibility constraints to maximize retained VMAF subject to a tunable perceptual-loss budget τ .

Paired trace-driven evaluation ($n=80$ episodes, identical policy and seeds) on FCC/Norway-style broadband traces shows that the VMAF-aware shield simultaneously improves mean QoE, rebuffering, and VMAF over legacy index-decrement projection, with all three gains statistically significant at $\tau=0.8$ (paired Wilcoxon signed-rank, $p<0.05$). A sweep of 47 shield configurations collapses to only 11 behavioral classes, identifying the VMAF-aware fallback as the dominant Pareto driver. Out-of-distribution replay on syn-

*Corresponding author

URL: psalehpour@tabrizu.ac.ir (Pedram Salehpour)

thetic 5G/mmWave stress traces corroborates significant QoE and rebuffer gains. Removing the shield at inference inflates rebuffering substantially ($p < 10^{-10}$), confirming that the policy and shield form a single deployable unit.

Keywords: Adaptive Bitrate Streaming, Deep Reinforcement Learning, Constrained MDP, Runtime Shielding, VMAF, Policy–Shield Co-design, Video QoE

1. Introduction

Client-side *adaptive bitrate* (ABR) policies trained with deep reinforcement learning (DRL) can achieve high average surrogate rewards, yet they frequently learn aggressive bitrate choices that concentrate catastrophic rebuffer risk in the tail [1, 2]. Sub-6 GHz and especially 5G/mmWave links routinely experience millisecond-to-second capacity cliffs caused by blockage, LOS/NLOS transitions, and mobility [3, 4]. These abrupt throughput drops convert rare dips into prolonged stalls that dominate viewer Quality of Experience (QoE), even when session-level averages appear acceptable [5, 6].

Production stacks therefore layer *runtime shields*—deterministic action projections executed before each chunk download—on top of learned policies [7]. The dominant legacy design is *content-blind*: when buffer-depletion risk triggers, the shield uniformly decrements the representation index until a viability test passes [1, 7]. Because perceptual fidelity is not affine in the discrete bitrate index, such decrements routinely over-pay in VMAF for modest safety gains: adjacent ladder rungs can differ by a few VMAF points or by ten, depending on content complexity and encoding [8]. Operators are thus forced into an opaque safety–quality trade-off, even when a feasible downgrade path with negligible perceptual loss exists on the same manifest [9, 10].

In this paper we reposition the shield as an *optimization layer* rather than a panic brake. We introduce a *VMAF-aware runtime shield*: a deterministic projection that, whenever an emergency step-down is required, searches the bitrate ladder under buffer-feasibility constraints to maximize retained VMAF subject to a tunable perceptual-loss budget τ and a catastrophic-ratio knob ρ_{cat} (Sec. 4.4). The projection is a transparent, auditable function of manifest chunk sizes, conservative throughput surrogates, buffer state, and per-chunk VMAF tables—not another opaque neural module [7, 8].

Paired trace-driven evaluation ($n=80$ episodes, identical policy and seeds)

demonstrates that the VMAF-aware shield *Pareto-improves* legacy index-decrement projection: at $\tau=0.8$, all three metrics—QoE, rebuffering, and VMAF—improve with statistical significance (paired Wilcoxon signed-rank, $p<0.05$; Sec. 5). Out-of-distribution replay on synthetic 5G/mmWave stress traces corroborates significant QoE and rebuffer gains (Sec. 5.9.1). Removing the shield at inference from a policy co-trained with shielding inflates rebuffering substantially, confirming that the policy and its shield form a single deployable unit [7].

The main contributions are:

- **CMDP formulation with Lagrangian PPO.** We cast ABR as a Constrained Markov Decision Process solved via primal–dual Lagrangian relaxation with PPO, manifest lookahead, and buffer-stabilization cues [11–13].
- **VMAF-aware runtime shielding.** We design a deterministic projection layer that optimizes emergency downgrade paths along the non-linear perceptual ladder, exposing two operator-facing knobs (τ , ρ_{cat}) without retraining (Sec. 4.4).
- **Paired statistical validation.** Wilcoxon signed-rank tests on $n=80$ paired episodes establish significant improvements in QoE, rebuffering, and VMAF at $\tau=0.8$ ($p<0.05$). Out-of-distribution 5G/mmWave stress replay confirms significant QoE and rebuffer gains (Secs. 5–5.9.1).
- **Policy–shield co-design.** We show that constrained policies learn proposals whose safety depends on the shield at inference, and that 47 shield configurations collapse to only 11 behavioral classes, with the VMAF-aware fallback as the dominant Pareto driver.

The remainder of this paper is organized as follows. Section 2 reviews related work; Section 3 formalizes the system model and CMDP; Section 4 details Lagrangian PPO and VMAF-aware projection; Section 5 presents the experimental evaluation; and Sections 6–7 discuss limitations and conclude.

2. Related Work

We organize the literature into three streams that converge on the research gap our VMAF-aware shield addresses. Our contribution targets

client-side bitrate selection under manifest and buffer observables—the control policy and its deterministic runtime projection—not throughput estimation. Data-driven predictors such as CS2P [14] are complementary infrastructure that any controller, including ours, may consume [1, 15]. Recent surveys [5, 16] and 5G/mmWave measurement studies [3] motivate why volatility-aware *control* matters independently of marginal gains in mean throughput forecasts.

2.1. Heuristic and model-based ABR

MPEG-DASH decouples packaging from transport; clients run local policies informed by ladder metadata and download observations [17]. Buffer-threshold heuristics such as BBA [18] achieve remarkable simplicity and transparency, while BOLA [19] provides Lyapunov-structured stability arguments for buffer evolution. RobustMPC [15] epitomizes predictor-assisted constrained lookahead, choosing bitrates by solving a short-horizon program under an estimated throughput trace.

Strengths. These designs are comparatively auditable: operators can reason about thresholds, horizons, and constraint violations without inspecting a neural graph.

Limitations. Hand-tuning does not scale across CDN regions, device classes, and blockage-heavy mmWave dynamics [3, 4]. When predictors lag abrupt capacity cliffs, even principled MPC objectives can misfire, and purely heuristic guards rarely exploit the nonlinear perceptual structure of the encoding ladder [8]. Recent SIGCOMM work on smoothness-aware on-line control (SODA [20]) sharpens production-relevant consistency objectives yet remains outside the DRL+shield co-design space we target. Orthogonal systems layers—edge assistance [21–23] and Oboe-style auto-tuning [24]—change *where* or *how* knobs are calibrated but do not resolve greedy neural tails or content-blind emergency projection.

2.2. Deep reinforcement learning for ABR

Pensieve [1] catalyzed trace-driven deep RL for ABR. Contemporaneous and follow-on DRL formulations such as D-DASH [25] demonstrate that neural policies can absorb heterogeneous observation histories and manifest features [2]. Recent PPO-based streaming RL [26] and hybrid predictor–learner pipelines [27] continue this trend.

Strengths. Scalarized RL objectives can track high average QoE surrogates on curated traces.

Critical tension. The policy network is a *black box*: maximizing expected return encourages aggressive bitrate choices whose failures concentrate in rare but catastrophic rebuffer tails under bursty cellular links [1, 6]. Field deployments (Fugu [6]) further highlight the distribution shift between simulators and live access networks. These limitations motivate CMDP and safe RL formulations in which declarative stall and smoothness budgets replace implicit reward scaling [9, 10].

2.3. Safe/constrained RL, shields, and the content-blindness gap

CMDPs and Lagrangian/primal–dual RL [11, 12, 28] supply the optimization vocabulary for constraint-aware training. Taxonomies of shields, Lyapunov methods, and risk-aware policy search can be found in [10, 29], and QoE-oriented safe RL studies increasingly import these ideas into streaming [9]. Runtime shielding [7] repairs or projects actions before execution, producing measurable intervention telemetry.

The unresolved gap. In practice, streaming shields overwhelmingly implement content-blind index decrements: when danger is detected, the client steps down the discrete ladder without considering how the nonlinear VMAF spacing amplifies perceptual damage [8]. Such shields trade opaque amounts of visual fidelity for safety, even when an alternative feasible downgrade path could preserve nearly all VMAF [9, 10].

Our departure. We treat the fallback itself as a perceptual optimization primitive: a deterministic VMAF-aware projection layer atop Lagrangian PPO, rather than another implicit QoE–stall compromise encoded in reward weights alone.

Summary. Unlike prior work that treats safety projection as a content-blind mechanism, we design the shield to *actively optimize* the emergency downgrade trajectory subject to buffer-feasibility tests, using manifest-known VMAF ladders [7, 8]. To our knowledge, this is the first trace-driven ABR architecture to embed nonlinear perceptual metrics inside a deterministic runtime shield and to demonstrate a strict empirical Pareto improvement (QoE, rebuffering, and VMAF, all paired) over legacy projection on identical policies and seeds (Sec. 5).

Table 1 summarizes the gap; Table 2 gives a finer-grained feature taxonomy (stall budgets, lookahead, neural map). Edge-assisted delivery [22, 23] and predictor innovations [14] stack orthogonally atop the control layer we focus on.

Table 1: High-level positioning for *client-side control*: deterministic pre-fetch projection (“Det. guard”) and whether emergency logic is **VMAF-ladder-aware** (not merely bitrate-index-aware). $\approx$ indicates constraint-aware training without a focal logged shield.

Representative work	Year	Paradigm	Det. guard	VMAF-aware
BBA [18]	2014	Heuristic	N	N
RobustMPC [15]	2015	MPC / lookahead	N	N
Pensieve [1]	2017	DRL	N	N
Oboe [24]	2018	Meta-tuning	N	N
Fugu / in-situ [6]	2020	Field DRL	N	N
SODA [20]	2024	Online control	N	N
QoE safe RL [9]	2025	Constr. DRL	$\approx$	N
Ours (Lag. PPO + VMAF shield)	—	CMDP + shield	Y	Y

Det. guard: deterministic action projection immediately before chunk download with auditable predicates (shield), distinct from stochastic policy sampling alone [7]. **VMAF-aware:** emergency search uses per-chunk VMAF ladder spacing, not only uniform index decrements [8].

Table 2: Finer-grained taxonomy of representative client-side ABR paradigms (complements the positioning grid of Table 1; symbol legend below).

Family (ref.)	Summarized objective / setting	Stall/ smooth budget	Manifest lookahead	Neural bitrate map	Runtime determ. guard
BBA [18]	Thresholds on buffer occupancy	Ind.	Imp. sizes	N	Heuristic thresh- olds
BOLA [19]	Online Lyapunov-structured bitrate utility	Imp. stabil- ity lens	Implicit (buffer cou- pling)	N	N
RobustMPC [15]	Short-horizon MPC under predicted throughput	Ind.	Horizon chunk sizes	N	N
Oboe [24]	System-level ABR parameter auto-tuning	N	N	N	N
Pensieve [1]	Scalar RL reward (QoE surrogate)	NF	Var. (blind- ing op- tional)	Y	N
D-DASH [25]	DRL on DASH traces	NF	Manifest ladder fea- tures	Y	N
Fugu / in-situ [6]	Field RL on live sessions	Imp.	Standard client ob- serv- ables	Y	N
QoE-aware safe RL [9]	Constrained / safe RL for streaming QoE	Focal	State design depend- ent	Y	Ctx
Hybrid control- learning [27]	Structured predictors plus learning	Var.	Often predictor- driven	P	N
Ours (Lag. PPO + optional shield)	Primal-dual CMDP surrogate trained with PPO [12, 13]	Adapt. dual λ	MLP policy + mani- fest LA [1, 17]	Y	Shield (opt.) [7]

Column symbols. **Y**=feature clearly present *in spirit*; **N**=not focal; **Ind.**=indirect via rewards/structure; **Imp.**=implicit through dynamics/buffer coupling; **NF**=no declarative constrained stall/smooth budget like ours; **Var.**=configuration-dependent; **P**=partial hybrid coupling; **Ctx**=context-dependent guarding. BOLA: analytic Lyapunov stabilization [19] (not neural CMDP).

3. System Model and Problem Formulation

3.1. Streaming architecture and control loop

We adopt the standard HTTP adaptive-streaming abstraction [1, 15, 17]. A *video origin/CDN* exposes a discrete bitrate ladder $\mathcal{B} = \{0, \dots, L-1\}$ with per-chunk sizes and perceptual metadata in the DASH/HLS manifest. The *client* maintains a playout buffer B_k and, at each chunk boundary k , requests the next segment over a time-varying bottleneck whose effective throughput $\hat{C}_k$ is exogenous to the controller in trace-driven emulation. On sub-6 GHz and especially mmWave access, $\hat{C}_k$ can jump within milliseconds due to blockage, LOS/NLOS transitions, and scheduling [3, 4]—the regime where greedy learned policies concentrate probability on tail stalls despite benign average traces [1, 6].

A neural policy π_θ maps an augmented state s_k to a stochastic *proposed* action $a_k^{\text{raw}} \in \mathcal{B}$ (sampled during training, argmax at evaluation). Before the download of chunk $k+1$ begins, the *VMAF-aware runtime shield* $\mathcal{S}$ applies a deterministic projection

$$(a_k^{\text{exec}}, \iota_k) = \mathcal{S}(s_k, a_k^{\text{raw}}), \quad a_k^{\text{exec}} \in \mathcal{B}, \quad \iota_k \in \{0, 1\}, \quad (1)$$

yielding the *executed* representation index a_k^{exec} and an audit bit ι_k that records whether projection altered the proposal. Only a_k^{exec} enters the chunk download and buffer dynamics (4); Figure 4.2 highlights the $a_k^{\text{raw}} \rightarrow \mathcal{S} \rightarrow a_k^{\text{exec}}$ path. Decisions evolve on the chunk clock with playback duration T_c per step (e.g., 4s in our harness) [1, 15].

Lagrangian PPO encodes stall and smoothness budgets as *soft* expectations learned over many rollouts: dual variables λ_r, λ_s reweight penalties, but a neural policy alone provides no per-step certificate that a dangerously large representation will never be attempted—especially under out-of-training throughput swings [3, 12, 28]. The shield supplies a complementary *hard*, deterministic repair: whatever π_θ proposes, the client executes a manifest-feasible a_k^{exec} that passes explicit buffer-margin viability tests [7].

3.2. State, action, reward, and nonlinear perceptual mapping

State s_k . Beyond classical telemetry (buffer occupancy, past throughput samples, previous decision), the client observation includes *manifest lookahead*: near-future chunk sizes across the ladder and spatial/temporal complexity tags (SI/TI), following MPC-style horizon inputs [1, 15, 17]. Crucially, the state also exposes the *per-chunk VMAF ladder* $\{V_k(j)\}_{j \in \mathcal{B}}$: for

chunk k , $V_k(j)$ is the perceptual score when representation j is fetched [8]. Without $\{V_k(j)\}$, neither the policy nor the shield could exploit content-dependent ladder spacing for emergency downgrade decisions.

Nonlinear, content-dependent fidelity. For each chunk k , the map $j \mapsto V_k(j)$ is neither affine in the discrete index j nor time-invariant across k : different GOP/texture structures yield different marginal VMAF gains between adjacent rungs. The composition “nominal bitrate $\rightarrow$ VMAF” is therefore a nonlinear, piecewise, content-varying mapping—precisely why legacy index-decrement shields can inflict disproportionate perceptual loss relative to a VMAF-aware search over the same feasible set [8].

Action. $a_k^{\text{raw}} \in \mathcal{B}$ is the policy’s proposed bitrate index for chunk $k+1$; the environment observes only a_k^{exec} .

QoE surrogate. Following RobustMPC/Pensieve reporting conventions [1, 15], we log a scalar per-chunk surrogate with fidelity $q_k \equiv V_k(a_k^{\text{exec}})$:

$$\text{QoE}_k^{(\text{rep})} = q_k - \alpha \text{smooth}_k - \beta \tau_k^{\text{stall}}, \quad (2)$$

where $\text{smooth}_k = |q_k - q_{k-1}|$ proxies quality switching, τ_k^{stall} is rebuffer seconds on step k , and α, β follow the shared evaluation harness. Session QoE is $\sum_{k \in \mathcal{K}_e} \text{QoE}_k^{(\text{rep})}$ (Section 5). The CMDP objective below separates primal fidelity $\sum q_k$ from explicit constraint channels on stall and smoothness statistics [11, 12].

3.3. Chunk-timescale dynamics

3.3.1. Throughput and latency

Let $\hat{C}_k > 0$ denote the effective throughput for chunk $k+1$ (trace-driven, comparable to baselines). Choosing a_k^{exec} induces payload size $\sigma_k(a_k^{\text{exec}})$ from the manifest; with $\tilde{C}_k = \max\{\hat{C}_k, \epsilon\}$, $\epsilon > 0$, nominal download seconds are $d_k = \sigma_k(a_k^{\text{exec}})/\tilde{C}_k$ [15].

3.3.2. Buffer and stall

Let $B_k \geq 0$ be buffer seconds at the start of download k [1, 15]. Rebuffering (stall) and buffer update follow

$$\tau_k^{\text{stall}} = \max\{0, d_k - B_k\}, \quad (3)$$

$$B_{k+1} = \max\{0, B_k - d_k\} + T_c. \quad (4)$$

Stochasticity enters through $\hat{C}_k$ on curated traces [16, 30, 31]; given $(s_k, a_k^{\text{exec}}, \hat{C}_k)$, transitions are deterministic.

3.4. CMDP costs, safety budgets, and Lagrangian relaxation

An unconstrained MDP maximizes $\mathbb{E}[\sum_k \gamma^k r_k]$ alone. We instead treat streaming as a *Constrained MDP* (CMDP): maximize expected fidelity subject to budgets on auxiliary costs induced by the same trajectory [11]. Let $c_k^{\text{stall}} \equiv \tau_k^{\text{stall}}$ denote the instantaneous stall cost (seconds of stall on step k) and $c_k^{\text{sm}} \equiv \text{smooth}_k$ the smoothness cost. A canonical discounted formulation requires, for discount $\gamma \in (0, 1]$ and budgets $d_{\text{rf}}, d_{\text{sm}}$,

$$\mathbb{E}_{\pi_{\theta}, \rho} \left[\sum_{k \in \mathcal{K}_e} \gamma^{k-1} c_k^{\text{stall}} \right] \leq d_{\text{rf}}, \quad \mathbb{E}_{\pi_{\theta}, \rho} \left[\sum_{k \in \mathcal{K}_e} \gamma^{k-1} c_k^{\text{sm}} \right] \leq d_{\text{sm}}, \quad (5)$$

In practice, we use undiscounted episodic summaries for logging and dual updates: the normalized stall ratio $\text{RB}(e)$ and the average smoothness cost $\text{SM}(e)$, both positive linear functionals of $\{c_k^{\text{stall}}\}$ and $\{c_k^{\text{sm}}\}$ over $\mathcal{K}_e = \{1, \dots, K_e\}$ [1, 12]. Let $T_e^{\text{play}} = K_e T_c$. Then

$$\text{RB}(e) = \frac{1}{T_e^{\text{play}}} \sum_{k \in \mathcal{K}_e} c_k^{\text{stall}}, \quad \text{SM}(e) = \frac{1}{K_e} \sum_{k \in \mathcal{K}_e} c_k^{\text{sm}}. \quad (6)$$

The constrained program motivating training is

$$\max_{\pi_{\theta}} \mathbb{E}_{\pi_{\theta}, \rho} \left[\sum_{k \in \mathcal{K}_e} q_k \right], \quad (7)$$

$$\text{s.t. } \mathbb{E}_{\pi_{\theta}, \rho} [\text{RB}(e)] \leq \delta_{\text{rf}}, \quad (8)$$

$$\mathbb{E}_{\pi_{\theta}, \rho} [\text{SM}(e)] \leq \delta_{\text{sm}}, \quad (9)$$

with targets $\delta_{\text{rf}}, \delta_{\text{sm}}$ fixed for reproducibility [15].

Directly enforcing (8)–(9) inside non-convex policy search is intractable; we therefore form the Lagrangian relaxation with per-step penalty channels $\widetilde{\text{RB}}_k, \widetilde{\text{SM}}_k$:

$$\mathcal{L}(\theta, \lambda) = \mathbb{E}_{\pi_{\theta}, \rho} \left[\sum_{k \in \mathcal{K}_e} (q_k - \lambda_r \widetilde{\text{RB}}_k - \lambda_s \widetilde{\text{SM}}_k) \right], \quad (10)$$

where $\lambda_r, \lambda_s \geq 0$ are adapted by projected dual ascent on empirical constraint gaps $g_r = \text{RB} - \delta_{\text{rf}}$, $g_s = \text{SM} - \delta_{\text{sm}}$ [11, 12]. The instantaneous reward used inside PPO reads as $q_k - \lambda_r \widetilde{\text{RB}}_k - \lambda_s \widetilde{\text{SM}}_k$ plus optional stabilizers (Sec. 4)—a standard “reward minus $\lambda \cdot \text{cost}$ ” construction. Finite-sample PPO with

clipped objectives does not certify constraint satisfaction at every iterate [13, 28]; this gap motivates the hard shield in (1).

Remark (auxiliary Lyapunov cues). Training optionally subtracts buffer-deviation and simulator Lyapunov penalties [32] outside (λ_r, λ_s) ; these serve as stabilizers, not formal certificates.

3.5. VMAF-aware shield as a hard projection

Let $\mathcal{A}_{\text{safe},k}(a_k^{\text{raw}}) \subseteq \mathcal{B}$ denote the set of representation indices that pass the shield’s buffer-margin and catastrophic-download viability tests at step k (Sec. 4.4), instantiated with pessimistic throughput, manifest sizes $\sigma_k(\cdot)$, and buffer B_k . *Legacy content-blind repair* maps a_k^{raw} downward by repeated index decrements until a feasible index is found [7].

VMAF-aware projection. Among feasible rungs, we select the one that maximizes perceptual score subject to a VMAF-loss budget τ relative to the proposal:

$$a_k^{\text{exec}} \in \arg \max_{j \in \mathcal{A}_{\text{safe},k}(a_k^{\text{raw}}) : V_k(a_k^{\text{raw}}) - V_k(j) \leq \tau} V_k(j), \quad (11)$$

with a strict light/strong fallback if (11) is empty [8]. Thus $\mathcal{S}$ is a deterministic perceptual projection $\pi_{\text{shield}} : (s_k, a_k^{\text{raw}}) \mapsto a_k^{\text{exec}}$ that optimizes retained VMAF on $\mathcal{A}_{\text{safe},k}$ instead of blindly stepping down the index order. Full predicate details (catastrophic ratio ρ_{cat} , pessimism ρ , margin μ) appear in Section 4.4.

4. Method: Lagrangian PPO and Runtime Shields

This section provides the implementation details needed to reproduce training and shielding. We build on Stable-Baselines 3 PPO [13, 33] with a custom Lagrangian reward wrapper and two deterministic shield variants: a QoE-oriented core (Algorithm 1) and a VMAF-aware perceptual projection (Algorithm 2). All symbols follow Section 3.

4.1. Observation interface and policy outputs

The state s_k and shield map $\mathcal{S}$ are defined in Section 3. The actor–critic stack consumes the observation tensors shown in Figure 4.2: throughput history, buffer level, last action, and manifest lookahead including SI/TI tags and per-chunk VMAF ladders [1, 8, 15, 17]. The actor outputs a categorical distribution over raw proposals $a_k^{\text{raw}} \in \mathcal{B}$. Optional hysteresis or shield-aware shaping wrappers may alter sampling incentives, but the executed action is always $a_k^{\text{exec}} = \mathcal{S}(s_k, a_k^{\text{raw}})$ when shielding is enabled [7].

4.2. Lagrangian PPO agent with Lyapunov-style buffer regularization

PPO backbone. We use standard clipped PPO [13] with twin Tanh MLP heads for $\pi_\theta(a \mid s_k)$ and $V_\phi(s_k)$, GAE $(\gamma, \lambda_{\text{GAE}})$, surrogate clipping, entropy bonuses, and Adam schedules (Table 3) [33].

CMDP shaping and stability cues. The train-time signal is the Lagrangian-relaxed objective (10) instantiated per step by (12). In addition to the dual-weighted stall and smoothness channels, we inject Lyapunov-style buffer regularizers $\phi(B_k)$ and emulator drift cues ψ_k^{Lya} with fixed coefficients (w_b, w_ℓ) outside the dual pair (λ_r, λ_s) [29, 32]. The regularizer ϕ penalizes dangerous buffer deviations early in training, tempering exploratory bitrate spikes before dual variables fully adapt. The drift cue ψ^{Lya} surfaces simulator-provided signals analogous to queue-stability heuristics. These terms are not formal safety certificates; they reduce variance and prevent premature dual divergence during trace-driven CMDP training [10, 12].

End-to-end control loop. The actor proposes a_k^{raw} ; the VMAF-aware shield $\mathcal{S}$ projects it to a_k^{exec} before the chunk download updates the client buffer under time-varying throughput. Manifest lookahead (including ladder VMAF tables) feeds the augmented state s_k ; auxiliary penalties $(\phi, \psi^{\text{Lya}})$ enter Lagrangian training only.

4.2.1. Lagrangian reward channel and dual ascent

Each step logs $(q_k, \text{rebuffer}_k, \text{smooth}_k, \phi(B_k), \psi_k^{\text{Lya}})$ into `info`, where $\widetilde{\text{RB}}_k, \widetilde{\text{SM}}_k$ correspond to the stall and smoothness penalties in (10). The PPO surrogate reward is (scaled by 1/100 for SB3 numerical stability)

$$r_k^{\text{train}} = \frac{1}{100} \left(q_k - \lambda_r \widetilde{\text{RB}}_k - \lambda_s \widetilde{\text{SM}}_k - w_b \phi(B_k) - w_\ell \psi_k^{\text{Lya}} \right). \quad (12)$$

After a warm-up phase ($H=60$ episodes), episodic statistics $\hat{\text{RB}}, \hat{\text{SM}}$ drive asymmetric projected dual ascent on constraint gaps $g_r = \hat{\text{RB}} - \delta_{\text{rf}}$, $g_s = \hat{\text{SM}} - \delta_{\text{sm}}$, with per-axis caps $[\lambda^{\min}, \lambda^{\max}]$ [10, 12]. Vectorized workers share the current $\lambda_\bullet$ values.

4.3. Catastrophic risk detection and pessimistic feasibility

The shields have no learnable parameters; each evaluates **deterministic** predicates on manifest data and a conservative throughput estimate. Let $\overline{\text{tp}}_k$ be the instantaneous trace throughput at chunk k and $\underline{C}_k$ the *pessimistic* estimate

$$\underline{C}_k = \max \left\{ C_{\min}, \rho \cdot \min \{ \overline{\text{tp}}_k, \text{last_tp}_k \} \right\}, \quad (13)$$

with pessimism $\rho \in (0, 1]$ and an emulator throughput floor $C_{\min}$. Default values are $\rho=0.97$ for training and $\rho=0.90$ for the VMAF-aware sweep reported in Section 5. For candidate representation j , predicted download seconds are

$$d_k^{\text{est}}(j) = \min\left\{\frac{\sigma_k(j)}{\underline{C}_k \cdot 10^3}, 60\right\}, \quad (14)$$

where $\sigma_k(j)$ is manifest chunk size in bits for bitrate index j at chunk k . A conservative estimate of the post-download buffer level (ignoring simultaneous playback credit during download) is

$$\tilde{B}_{k+1}(j) = \max\{0, B_k - d_k^{\text{est}}(j)\} + T_c, \quad (15)$$

Operators can compare $\tilde{B}_{k+1}(j)$ against a safety floor B_{safe} ; if it falls below this threshold, representation j is deemed emergency-infeasible.

Risk gate. When enabled, the shield passes through a_k^{raw} unchanged whenever

$$d_k^{\text{est}}(a_k^{\text{raw}}) \leq \max\{B_k, 0.1\} \cdot \rho_{\text{risk}}, \quad (16)$$

with $\rho_{\text{risk}}=1.35$ as default; benign proposals therefore bypass projection [7].

Catastrophic pass-through. In the VMAF-aware path, if $d_k^{\text{est}}(a_k^{\text{raw}}) \leq B_k \rho_{\text{cat}}$, the shield returns $a_k^{\text{exec}} = a_k^{\text{raw}}$ without constructing the VMAF candidate pool (Algorithm 2). The **light** mode uses the same inequality as a fast no-intervention check before margin-based repair.

Critical buffer. If $B_k \leq B_{\text{crit}}$ (default 0.20–0.30 s), the shield snaps to index 0 as a hard emergency rule.

4.4. Deterministic runtime projection

Role. $\mathcal{S} : (s_k, a_k^{\text{raw}}) \mapsto (a_k^{\text{exec}}, \iota_k)$ executes *before* `env.step`; $\iota_k \in \{0, 1\}$ logs interventions for audit CSVs (Section 5).

4.4.1. QoE-oriented shield core

Besides **off**, the shield exposes three operating modes: **qoe**, **light**, and **strong**. All modes share $\underline{C}_k$ and $d_k^{\text{est}}(\cdot)$ from (13)–(14), clip a_k^{raw} to $\mathcal{B}$, apply the critical-buffer snap, and honor the optional risk gate (16). The **qoe** mode searches downward from a_k^{raw} over at most Δ_{dn} index steps (default 2), but *accepts* representations whose predicted download fits a small **predicted-stall budget** ζ (default 0.25 s), i.e. $d_k^{\text{est}}(j) \leq B_k + \zeta$, before widening search down to a minimum guard index $j_{\min}$ (default 1). If no candidate satisfies the budget, the shield may emit an emergency index 0 when the minimum guard

index $j_{\min}$ would imply a predicted stall above threshold ξ (default 1.25 s); otherwise it returns $j_{\min}$. The **light** and **strong** modes apply buffer-margin feasibility tests with defaults $\rho_{\text{cat}}=2.50$, $\mu_{\ell}=0.15$ s, $\mu_s=0.75$ s. Algorithm 1 details the full procedure; the shield wrapper may additionally cap post-intervention upshifts during a short recovery window.

4.4.2. VMAF-aware perceptual projection

Legacy path. Without VMAF awareness, the **light** and **strong** modes perform index-decrement search with margins μ . **VMAF-aware path.** Define the soft-safe candidate set (never upswitching)

$$\mathcal{J}_k = \left\{ j \in \mathcal{B} : j \leq a_k^{\text{raw}}, d_k^{\text{est}}(j) \leq B_k \sigma_{\text{soft}} \right\}, \quad (17)$$

where σ_{soft} (**soft_tolerance**) relaxes the hard buffer equality into a tunable slack band. Let $V_k(j)$ be manifest VMAF for index j at chunk k (monotone proxy if missing). Among candidates whose perceptual loss vs. the proposal stays within budget τ (**vmaf_loss_budget**),

$$\mathcal{J}'_k = \left\{ j \in \mathcal{J}_k : V_k(a_k^{\text{raw}}) - V_k(j) \leq \tau \right\}, \quad (18)$$

pick $\mathcal{J}_k^* = \mathcal{J}'_k$ if non-empty, otherwise $\mathcal{J}_k^* = \mathcal{J}_k$. The implementation returns

$$a_k^{\text{exec}} = \arg \max_{j \in \mathcal{J}_k^*} \left(V_k(j), j, -d_k^{\text{est}}(j) \right), \quad (19)$$

i.e. maximize VMAF, break ties toward the higher index, then shorter predicted download (lexicographic tie-breaking). If $\mathcal{J}_k = \emptyset$, the shield falls back to index 0. **Lookahead gate (optional).** When a lookahead horizon $H > 0$ is configured, a short pessimistic rollout under the raw action can bypass projection if buffers never dip below a minimum threshold; this gate is evaluated *before* the VMAF-aware branch.

Design-space remark. Sweeps over 47 configurations collapse to 11 behavioral classes dominated by whether VMAF-aware fallback is enabled (Section 5.7).

4.5. Optional shield-aware shaping and hysteresis

A shield-aware penalty wrapper penalizes shield interventions: $\beta_{\text{sh}} \iota_k + \gamma_{\text{sh}} |a_k^{\text{exec}} - a_k^{\text{raw}}|$ with $\beta_{\text{sh}}=0.08, \gamma_{\text{sh}}=0.03$. A hysteresis wrapper caps index swings and blocks upswitches when $B_k < 1.5$ s. The shield wrapper optionally applies smooth-recovery upshift caps after interventions.

4.6. End-to-end training loop (primal-dual PPO + optional shield)

Algorithm 3 composes wrappers with SB3-PPO; Table 3 freezes hyperparameters.

4.7. Policy-shield co-design strategy

Lagrangian PPO enforces CMDP budgets only in **expectation** over roll-outs: λ_r, λ_s chase $\delta_{\text{rf}}, \delta_{\text{sm}}$, but the neural map π_θ can still propose aggressive representations on individual steps—especially under 5G/mmWave draws outside the training mixture [3, 12]. The shield is the complementary **deterministic safety net**: it does not replace π_θ ’s representational capacity but repairs individual proposals before they reach the plant [7].

Co-design. When shields wrap training, π_θ learns to rely on this safety net. The QoE-oriented guard (Algorithm 1) pairs with Lagrangian PPO during training, while the VMAF-aware perceptual projection (Algorithm 2) constrains catastrophic downloads under explicit VMAF budgets at inference [8]. Removing the shield at inference breaks the implicit contract and inflates rebuffering (Section 5.8). The observed Pareto improvement is a property of the *joint* system (Lagrangian PPO + deterministic shield), not of either module in isolation. We recommend shipping the same shield configuration used in training and tuning the perceptual knobs $(\tau, \rho_{\text{cat}})$ at deployment rather than disabling guards.

4.8. Implementation notes and hyperparameters

The implementation consists of Lagrangian reward wrappers and deterministic shield modules. Table 3 summarizes the PPO and Lagrangian hyperparameters; remaining defaults ($\delta_{\text{rf}}, \delta_{\text{sm}}$, dual step sizes, warm-up length, and shield ratios) are documented in the reproducibility repository.

Table 3: PPO and Lagrangian hyperparameters (evaluation freeze).

Item	Setting
Policy / value MLP	Two Tanh layers, width 256 each [13]
Rollout horizon	$4096 \cdot 8$ parallel env steps per PPO iteration
Learning rate schedule	Piecewise-linear decay $3 \times 10^{-4} \rightarrow 10^{-5}$
Advantage	GAE ($\gamma=0.99, \lambda_{\text{GAE}}=0.95$)
Constraint wrapper	Projected asymmetric dual ascent on $(\hat{\text{R}}\text{B}, \hat{\text{S}}\text{M})$; warm-up episodes
Entropy coef (proposed)	0.04 (ablations lowered)
Reward scaling	Divide shaped reward by 100 for SB3 numerical range
Shield (reported sweep)	VMAF-aware projection; levels light / strong ; risk gate optional
Throughput pessimism ρ	0.90 (VMAF-aware sweep); 0.97 (training; Sec. 4.3)

Algorithm 1 QoE-oriented shield core

```
1: Input:  $a^{\text{raw}}$ , buffer  $B_k$ , chunk  $k$ , trace + last_tp,  $\sigma_k(\cdot)$ ; cfg  
   (level,  $B_{\text{crit}}$ ,  $j_{\text{min}}$ ,  $\rho$ ,  $\rho_{\text{risk}}$ ,  $\rho_{\text{cat}}$ ,  $\Delta_{\text{dn}}$ ,  $\zeta$ ,  $\xi$ ,  $\mu_\ell$ ,  $\mu_s$ , risk_on, emergency)  
2:  $a \leftarrow \text{clip}(a^{\text{raw}}, 0, L-1)$ ; compute  $\underline{C}_k$ ,  $d_k^{\text{est}}(\cdot)$  via (13)–(14)  
3: if level = off then  
4:   return ( $a$ , 0)  
5: end if  
6: if  $B_k \leq B_{\text{crit}}$  then  
7:   if  $a = 0$  then  
8:     return (0, 0)  
9:   else  
10:    return (0, 1)  
11:   end if  
12: end if  
13: if risk_on and  $d_k^{\text{est}}(a) \leq \max\{B_k, 0.1\} \rho_{\text{risk}}$  then  
14:   return ( $a$ , 0)  
15: end if  
16: if level = qoe then  
17:    $f \leftarrow \max\{j_{\text{min}}, a - \Delta_{\text{dn}}\}$   
18:   for  $c = a, a-1, \dots, f$  do  
19:     if  $d_k^{\text{est}}(c) \leq B_k + \zeta$  then  
20:       if  $c = a$  then  
21:         return ( $c$ , 0)  
22:       else  
23:         return ( $c$ , 1)  
24:       end if  
25:     end if  
26:   end for  
27:   for  $c = f-1, f-2, \dots, j_{\text{min}}$  do  
28:     if  $d_k^{\text{est}}(c) \leq B_k + \zeta$  then  
29:       if  $c = a$  then  
30:         return ( $c$ , 0)  
31:       else  
32:         return ( $c$ , 1)  
33:       end if  
34:     end if  
35:   end for  
36:   if emergency and  $j_{\text{min}} > 0$  and  $\max\{0, d_k^{\text{est}}(j_{\text{min}}) - B_k\} > \xi$  then  
37:     if  $a = 0$  then  
38:       return (0, 0)  
39:     else  
40:       return (0, 1)  
41:     end if  
42:   end if  
43:   if  $a = j_{\text{min}}$  then  
44:     return ( $j_{\text{min}}$ , 0)
```

Algorithm 2 VMAF-aware perceptual projection

```

1: Input: raw index  $a^{\text{raw}}$ , buffer  $B_k$ , chunk  $k$ , trace + last throughput, bits
    $\sigma_k(\cdot)$ , VMAF table  $\{V_k(j)\}_{j \in \mathcal{B}}$ ,  $(\rho, \rho_{\text{cat}}, \rho_{\text{risk}}, \sigma_{\text{soft}}, \tau, B_{\text{crit}})$  (after optional
   lookahead gate)
2:  $a \leftarrow \text{clip}(a^{\text{raw}}, 0, L-1)$ ; compute  $\underline{C}_k$  by (13);  $\forall j : d_k^{\text{est}}(j)$  by (14)
3: if  $B_k \leq B_{\text{crit}}$  then
4:   if  $a = 0$  then
5:     return  $(0, 0)$ 
6:   else
7:     return  $(0, 1)$ 
8:   end if
9: end if
10: if risk gate enabled and  $d_k^{\text{est}}(a) \leq \max\{B_k, 0.1\}\rho_{\text{risk}}$  then
11:   return  $(a, 0)$ 
12: end if
13: if not VMAF-aware mode then
14:   apply legacy light/strong index projection; return
15: end if
16: if  $d_k^{\text{est}}(a) \leq B_k \rho_{\text{cat}}$  then
17:   return  $(a, 0)$ 
18: end if
19: build  $\mathcal{J}_k$  per (17)
20: if  $\mathcal{J}_k = \emptyset$  then
21:   if  $a = 0$  then
22:     return  $(0, 0)$ 
23:   else
24:     return  $(0, 1)$ 
25:   end if
26: end if
27:  $\mathcal{J}'_k \leftarrow \{j \in \mathcal{J}_k : V_k(a) - V_k(j) \leq \tau\}$ ;  $\mathcal{J}^*_k \leftarrow \mathcal{J}'_k$  if non-empty else  $\mathcal{J}_k$ 
28:  $a^{\text{exec}} \leftarrow \arg \max_{j \in \mathcal{J}^*_k} (V_k(j), j, -d_k^{\text{est}}(j))$  (lexicographic)
29: if  $a^{\text{exec}} = a$  then
30:   return  $(a^{\text{exec}}, 0)$ 
31: else
32:   return  $(a^{\text{exec}}, 1)$ 
33: end if

```

Algorithm 3 Lagrangian PPO training with optional shield-at-train

```

1: Instantiate vectorized SB3-PPO( $\theta$ ); wrap envs with
   LagrangianRewardWrapper lineage and optional hysteresis /
   ShieldAwarePenalty.
2: for optimization iteration until budget do
3:   for parallel worker  $w$  do
4:     Reset env; load shared  $(\lambda_r, \lambda_s)$  per artifact.
5:     for environment step  $k=1:K_e$  do
6:        $a_k^{\text{raw}} \sim \pi_\theta(\cdot \mid s_k)$ 
7:        $a_k^{\text{exec}} \leftarrow \mathcal{S}(s_k, a_k^{\text{raw}})$  if shield wraps training env, else  $a_k^{\text{raw}}$ 
8:       Step emulator with  $a_k^{\text{exec}}$ ; log  $\{q_k, \widetilde{\text{RB}}_k, \widetilde{\text{SM}}_k, \phi, \psi^{\text{Lya}}\}$ 
9:        $r_k^{\text{train}} \leftarrow$  Eq. (12) plus optional shield-aware shaping
10:    end for
11:    Episode end: update moving averages of  $\hat{\text{RB}}, \hat{\text{SM}}$ ; if past warm-up
        $\Rightarrow$  projected dual ascent  $\Pi_\Lambda$ 
12:    Push transitions into PPO rollout buffer
13:  end for
14:  PPO.update() (GAE + clipped surrogate + entropy)
15: end for

```

5. Experimental Evaluation

We organize the evaluation around five research questions:

- RQ1** *Pareto front.* Does the VMAF-aware fallback Pareto-dominate uniform-decrement projection at matched rebuffer budgets?
- RQ2** *Baseline competitiveness.* Where do the recommended operating points sit relative to RobustMPC and Pensieve?
- RQ3** *Component ablations.* What is the isolated value of manifest lookahead, Lyapunov cues, and Lagrangian dual updates?
- RQ4** *Shield design-space structure.* Across 47 shield variants, do behaviors collapse into a small number of equivalence classes?
- RQ5** *Policy-shield co-design.* To what extent do constrained-trained policies depend on the shield at inference?

5.1. Experimental setup and datasets

We evaluate on a multi-video trace-driven emulator ($K_e=48$ chunks per episode, fixed segment duration T_c as in Sec. 3).

Primary suite (broadband). FCC/Norway-style partitioned throughput traces from public community releases [30, 31]; four videos $\times$ twenty paired seeds per method after warmup.

Stress suite (OOD). Synthetic 5G/mmWave-style traces with intermittent LOS/NLOS segments and outage plateaus that stress tail risk [3]. Section 5.9.1 reports paired replay of the *same* constrained policy under identical harness assumptions.

Baselines. Pensieve is reproduced with observation masking for fairness; RobustMPC follows [15]. Genie [1] uses non-causal trace knowledge and serves only as an oracle reference.

Shield pairing protocol. All legacy vs. VMAF-aware comparisons hold the trained policy, ladder, traces, and Video $\times$ Episode identifiers fixed and vary only the inference-time shield logic, enabling per-episode paired tests.

Table 4: Baselines vs. same-policy shield ($n=80$). Lower block is paired; absolute QoE not comparable across blocks.

Method	QoE	VMAF	Rebuf. (%)	Intv. (%)
<i>Baselines</i> (distinct policies; 95% CI, $n=80$)				
Pensieve [1]	3269 [2931, 3573]	71.59 [65.03, 77.79]	14.99 [12.29, 17.86]	—
RobustMPC [15]	3275 [2953, 3577]	74.65 [68.40, 80.38]	30.70 [27.28, 34.10]	—
<i>Same CMDP policy</i> (paired replay; CI)				
+ Legacy (index decrement)	2518.87 [2176.5, 2854.7]	65.82	5.67 [4.95, 6.36]	3.23
+ VMAF-aware ($\tau=1.0$)	2561.00 [2216.3, 2900.5]	66.04	5.12 [4.41, 5.84]	2.71

Table 5: Extended policy-mode means with bootstrap 95% CIs ($n=80$). Format: *mean* [lo, hi].

Method	QoE $\uparrow$	VMAF $\uparrow$	Rebuf. (%) $\downarrow$	Switches $\downarrow$	Type
Genie (oracle) [1]	3365 [3043, 3663]	72.15 [65.44, 78.26]	8.28 [6.02, 10.95]	3.7 [3.2, 4.1]	oracle
RobustMPC [15]	3275 [2953, 3577]	74.65 [68.40, 80.38]	30.70 [27.28, 34.10]	8.4 [7.7, 9.1]	MPC
Pensieve [1]	3269 [2931, 3573]	71.59 [65.03, 77.79]	14.99 [12.29, 17.86]	6.4 [5.6, 7.2]	DRL
Fugu [6]	3259 [2924, 3558]	74.05 [67.36, 80.05]	20.12 [17.75, 22.63]	14.9 [13.7, 16.1]	IL
BBA [18]	3054 [2718, 3373]	72.89 [66.62, 79.11]	27.92 [24.35, 31.43]	25.6 [24.2, 27.0]	heur.
Ours: CMDP (no shield)	3270 [2945, 3569]	71.65 [65.06, 78.14]	14.73 [11.73, 17.78]	6.5 [5.7, 7.3]	RL+CMDP
Ours: + shield	3218 [2889, 3529]	68.47 [61.91, 75.05]	1.45 [1.06, 1.86]	5.2 [4.5, 6.0]	RL+CMDP+S
Ours: + risk-gate	3225 [2880, 3536]	68.61 [61.95, 75.07]	1.20 [0.85, 1.60]	6.3 [5.2, 7.5]	RL+CMDP+RG
Ours: + shield-QoE †	3224 [2896, 3536]	68.74 [61.87, 75.14]	1.45 [1.07, 1.87]	9.9 [8.7, 11.4]	RL+CMDP+Q

Interval: bootstrap 95% CI on episode means ($n=80$ per row). † Shield-aware variant (elevated bitrate switches).

5.2. Performance headline and baselines

Table 4 summarizes the primary comparison: classical baselines (distinct policies) versus the same constrained policy under legacy and VMAF-aware projection. Absolute QoE levels are *not* comparable across the two partitions (different policies and surrogate bookkeeping); within the lower partition, every shift is paired on identical seeds. Extended results including Genie and ablated agents appear in Table 5.

5.3. Overall performance (extended means)

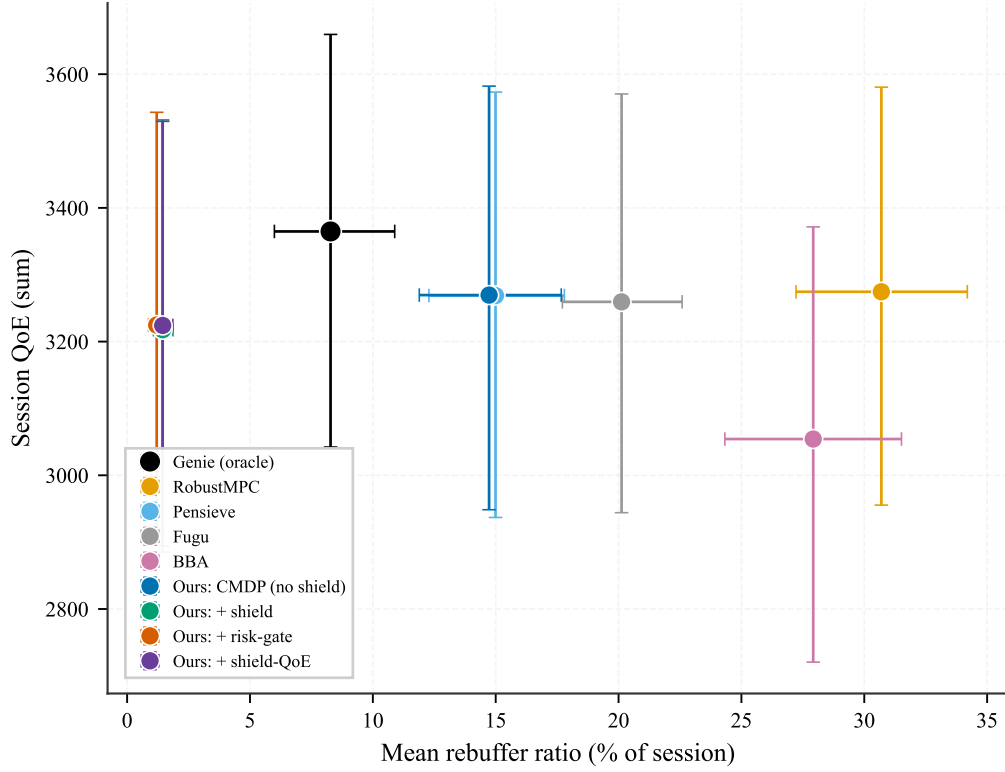


Figure 1: Safety-QoE trade-off ($n=80$ per method): session QoE vs. mean rebuffer ratio with bootstrap 95% CIs.

Table 6: Paired Wilcoxon signed-rank tests on QoE vs. Pensieve and RobustMPC (two-sided p).

Method	vs. Pensieve		vs. RobustMPC	
	med. Δ QoE	p	med. Δ QoE	p
Ours: CMDP (no shield)	6.0	0.259	8.6	0.124
Ours: + shield	-59.4	3.42×10^{-10}	-55.7	1.34×10^{-9}
Ours: + risk-gate	-42.6	7.07×10^{-7}	-37.9	6.90×10^{-7}
Ours: + shield-QoE [†]	-35.3	4.94×10^{-8}	-40.9	7.31×10^{-6}

[†]Two-sided paired Wilcoxon signed-rank; Δ QoE is the per-episode paired difference (method minus baseline).

Table 7: Paired Wilcoxon signed-rank tests on rebuffer ratio (%) vs. Pensieve and RobustMPC.

Method	vs. Pensieve		vs. RobustMPC	
	med. Δ Rebuf. (%)	p	med. Δ Rebuf. (%)	
Ours: CMDP (no shield)	−0.11	0.310	−11.11	2
Ours: + shield	−10.25	4.58×10^{-14}	−29.86	7
Ours: + risk-gate	−10.39	4.23×10^{-14}	−29.86	7
Ours: + shield-QoE [†]	−10.73	7.90×10^{-14}	−30.23	7

Same paired episodes; Δ Rebuffer is the per-episode difference in reported mean rebuffer ratio (%). Negative values

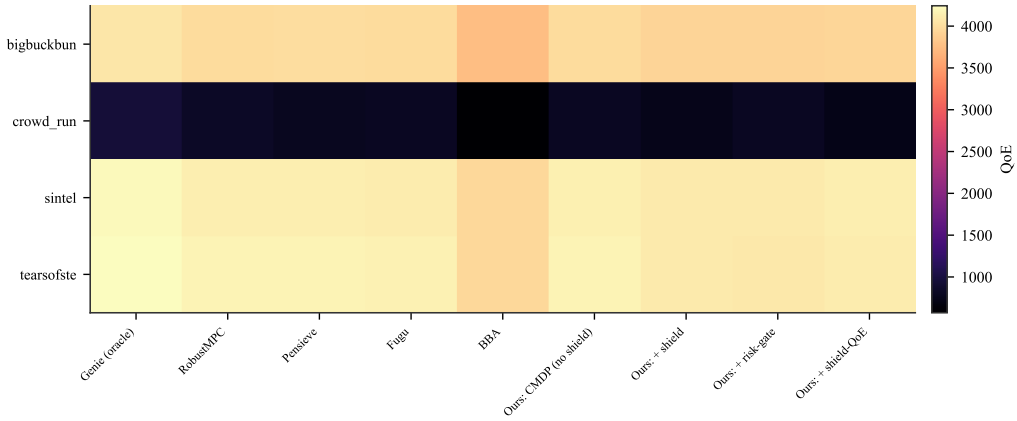


Figure 2: Mean QoE per video across all methods. Content-dependent variation is evident: motion-heavy sequences (e.g., **crowd_run**) lower QoE uniformly.

Table 8: Paired Wilcoxon signed-rank: VMAF-aware ($\tau=1.0$) vs. legacy projection ($n=80$). QoE and rebuffer improve significantly ($p<0.01$).

Metric	Mean Δ	Med. Δ	p
QoE (target – legacy)	+42.13	0.00	0.0050
Rebuffer (% pts)	−0.55	0.00	0.0029
VMAF	+0.22	0.00	0.0564

5.4. Statistical significance (paired shield vs. legacy)

Means can hide heavy tails; paired nonparametric tests anchor claims about *systematic* shifts on identical episodes. Table 8 reports Wilcoxon signed-rank outcomes for the recommended VMAF-aware operating point ($\tau=1.0$, budget 8) against legacy projection on $n=80$ paired traces.

Result. QoE and rebuffering improve with $p<0.01$. VMAF at $\tau=1.0$ is marginally significant ($p=0.056$), while tightening to $\tau=0.8$ yields a strictly significant triple at $\alpha=0.05$ (VMAF $p=0.041$; Table 9).

RQ1: Pareto improvement over legacy. Tables 8 and 9 jointly answer RQ1. At $\tau=1.0$, the VMAF-aware shield significantly improves QoE and rebuffering ($p<0.01$) with a marginal VMAF gain ($p=0.056$). Tightening to $\tau=0.8$ yields a strictly significant triple ($p<0.05$ on all three metrics), constituting a strict empirical Pareto improvement. Section 5.9.1 extends this analysis to synthetic 5G/mmWave traces, where QoE and rebuffer gains remain highly significant while VMAF is not reliably separated—an expected artifact when sessions become stall-saturated (Table 10).

Table 9 reports the full sweep. The VMAF-aware family (rows 2–5) Pareto-dominates the legacy anchor on QoE, rebuffering, *and* VMAF concurrently. By contrast, threshold-only sweeps that vary ρ_{cat} *without* the VMAF-aware fallback move QoE up by smaller, statistically non-significant margins while *worsening* rebuffering by 2–5 percentage points (Section 5.7). This isolates the VMAF-aware fallback as the active Pareto driver: it is the *shape* of the projection rule, not the threshold itself, that bends the front.

RQ2: Cross-baseline context. Under the same evaluation harness, the recommended VMAF-aware operating point reduces mean session rebuffering by 9.87 % versus Pensieve (14.99 %) and by 25.58 % versus RobustMPC (30.70 %). Absolute QoE numbers are not directly paired across distinct trained policies; we therefore report them descriptively in Tables 5–7 rather than as a strict statistical claim. Figure 3 provides a forest-plot visualization of paired QoE deltas.

Table 9: Shield sweep (same policy, seeds, $n=80$). VMAF-aware variants Pareto-dominate legacy on QoE, rebuffer, and VMAF; threshold-only variants do not.

Variant	QoE	Reb. (%)	VMAF	Int. (%)
Legacy ($\rho_{\text{cat}}=2$, no VMAF FB)				
<i>VMAF-aware</i>				
$\tau=0.8$, bud. 8				
$\tau=1.0$, bud. 8		5.12		
$\tau=1.2$				—
$\tau=1.5$				—
<i>Threshold only (no VMAF FB)</i>				
$\rho_{\text{cat}}=3$			—	—
$\rho_{\text{cat}}=4$			—	—
$\rho_{\text{cat}}=5$			—	—
<i>Threshold + VMAF FB</i>				
$\rho_{\text{cat}}=3$				—
$\rho_{\text{cat}}=4$				—
<i>Shield off</i>				
Shield off		11.04		0.00

5.5. Tail behavior: empirical CDFs

Means and CIs summarize central tendency but obscure tails—precisely where shields must prove their value. Figures 4–5 plot session QoE and rebuffer ratio CDFs under the *policy-mode* harness (distinct learned controllers, Table 5). Figures 6–7 isolate the *paired* replay: the identical constrained policy with shield off, legacy projection, and VMAF-aware projection ($\tau=1.0$). On QoE, the VMAF-aware curve lies to the right of legacy (higher is better); on rebuffer, it rises faster toward unit mass near zero stall—consistent with the Pareto shifts in Table 9.

5.6. Pareto dominance visualizations

Figures 8–9 summarize the shield sweep in the QoE–rebuffer plane and mark the empirical non-dominated set; Figure 10 shows per-episode paired ΔQoE versus legacy.

Statistical reporting. Tables 6–7 excerpt paired Wilcoxon outcomes; full grids for every shield variant are exported as CSV in the reproducibility repository. The VMAF-aware-vs-legacy comparison (Table 9, rows 2–3 vs.

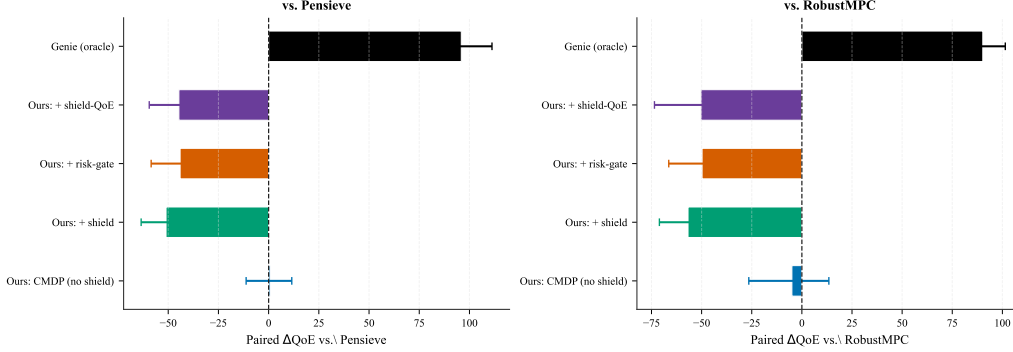


Figure 3: Forest plot of paired Δ QoE versus Pensieve (left) and RobustMPC (right) with bootstrap 95% CIs per video and overall.

row 1) is the primary headline: at $\tau=0.8$, paired Δ QoE, Δ rebuffer, and Δ VMAF are all statistically significant; at $\tau=1.0$, QoE and rebuffer remain highly significant while VMAF is marginal ($p \approx 0.056$). For distributional-shift scenarios with extreme outage-driven rebuffer (Sec. 5.9.1), metrics should be interpreted dimension-wise: paired QoE and stall shifts remain primary evidence, while perceptual deltas may require larger n or equivalence-style tests [34] for VMAF non-inferiority claims.

5.7. Shield-design-space collapse (RQ4)

Sweep design. We instantiate 47 shield configurations spanning four axes: (i) catastrophic ratio $\rho_{\text{cat}} \in \{2, 3, 4, 5, \infty\}$; (ii) VMAF-aware fallback on/off with VMAF-loss budget $\tau \in \{1, 4, 8\}$; (iii) H -step lookahead pre-gating with $H \in \{1, 2, 3\}$ and minimum-buffer thresholds; and (iv) risk-gate ratio ρ_{risk} . Every configuration is evaluated on the same 80 paired episodes.

Behavioral collapse. Despite spanning a 47-point grid, the resulting (QoE, Rebuffer, VMAF) outcomes cluster into only 11 distinct behavioral classes (grouped at ± 5 QoE / $\pm 0.3\%$ rebuffer tolerance). The dominant Pareto driver is *whether the VMAF-aware fallback is enabled*: variants in the same VMAF-tolerance band collapse to identical operating points regardless of secondary tweaks (lookahead horizon, minimum-buffer threshold, throughput-scale parameter, risk-gate ratio). Lookahead pre-gating contributes ≤ 1.5 QoE points once VMAF-fallback and ρ_{cat} are fixed, and risk-gating influences mainly the intervention rate rather than aggregate QoE or rebuffering.

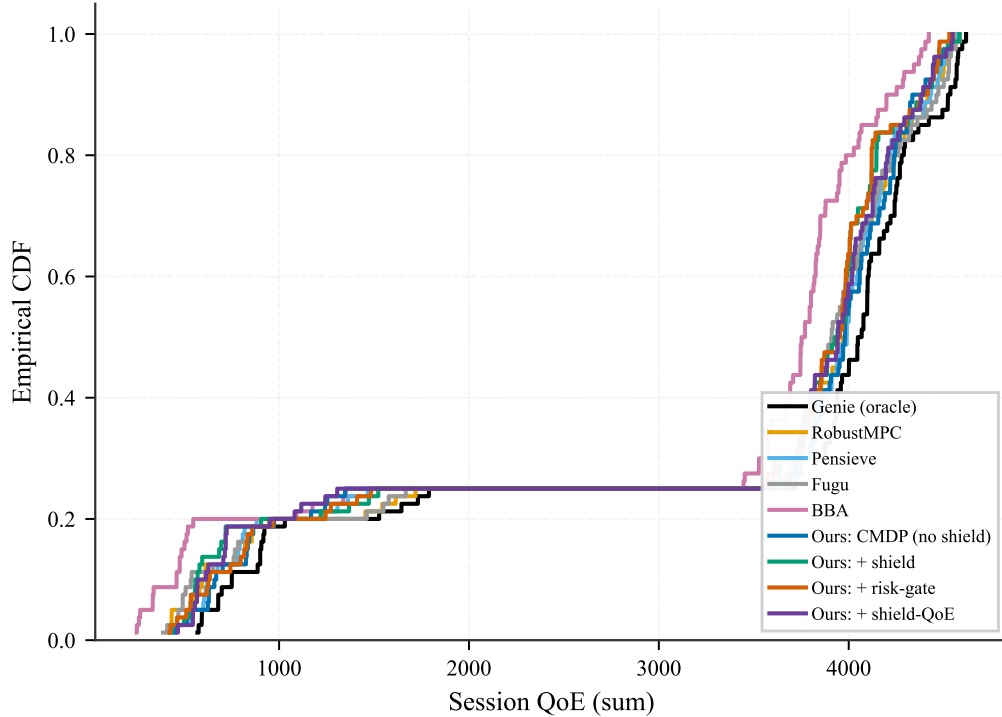


Figure 4: Empirical CDF of session QoE under policy-mode evaluation (Table 5). Right-shifted curves indicate higher QoE.

VMAF-aware fallback as the active Pareto driver. A direct comparison at $\rho_{\text{cat}}=3$ with and without VMAF-fallback (Table 9) shows that the fallback produces +33 QoE and -0.74 percentage-point rebuffering simultaneously while only marginally raising the intervention rate, isolating projection *shape* from threshold *level*. The same pattern holds at $\rho_{\text{cat}}=4$. We therefore recommend treating the VMAF-aware fallback as the primary architectural choice and exposing τ and ρ_{cat} as the two operator-facing deployment knobs; the remaining parameters can be fixed at sensible defaults without measurable Pareto cost.

5.8. Policy-shield co-design (RQ5)

Setup. Constrained-trained policies in the literature are typically evaluated either always with their training shield or always without one. We instead evaluate the same trained policy under both regimes on identical

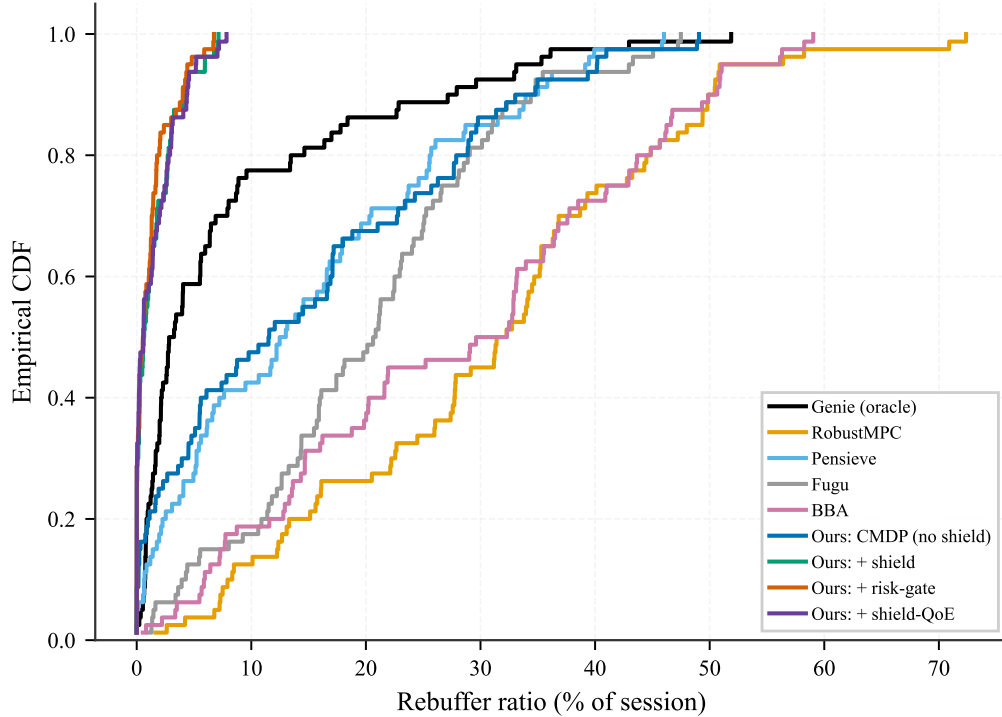


Figure 5: Empirical CDF of rebuffer ratio (% of session). Curves rising steeply near zero indicate less stalling.

traces.

Result. Removing the shield at inference more than doubles mean rebuffering (from 5.12 % with the recommended VMAF-aware shield to 11.04 % unshielded), a paired -5.92 % drop with $p=1.4 \times 10^{-10}$ ($n=80$). QoE moves only marginally (+2.4 paired, not significant). This indicates that the policy learns to propose bitrate sequences whose safety is conditional on downstream projection: the shield is not an optional tail-risk mitigator but an operationally essential component.

Implication. We advocate *co-design* (train with shield, ship with shield) as the operationally faithful deployment pattern, with τ and ρ_{cat} —rather than shield removal—as the deployment-time tunables. Removing the shield to “unbias” a policy comparison silently moves the controller off every Pareto point in Table 9 and substantially degrades rebuffering; any A/B comparison should treat the policy and shield as a single deployable unit.

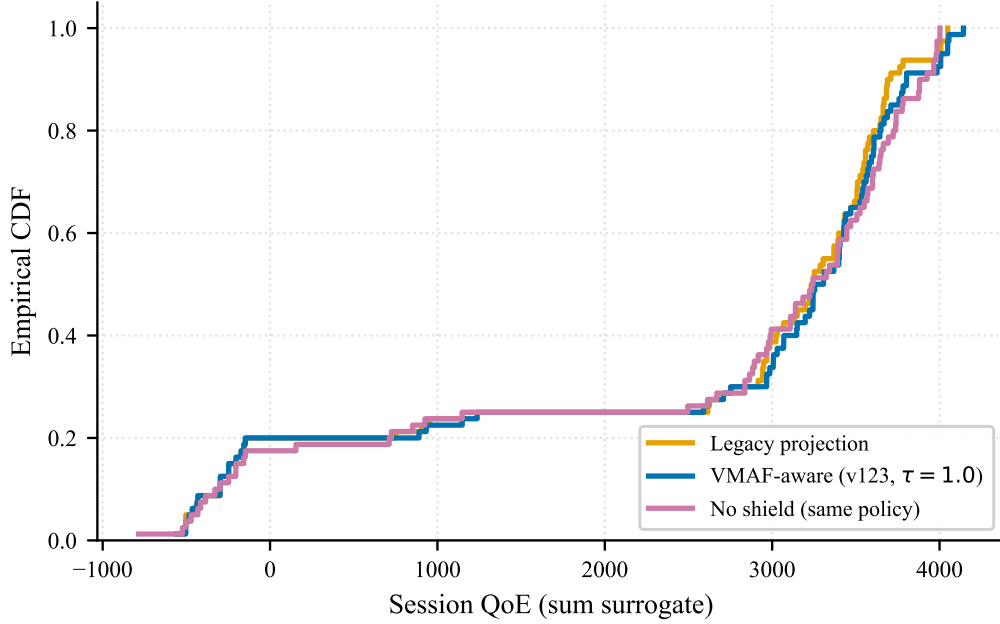


Figure 6: Paired replay CDF of session QoE ($n=80$): identical policy, traces, and seeds; only the shield logic varies.

5.9. Generalization across heterogeneous networks

Broadband vs. volatile cellular. On the primary FCC/Norway-style suite, throughput partitions are comparatively stable: the constrained policy sits near Pensieve-class QoE and the shield family primarily *repairs* rare risky proposals rather than fighting persistent stall (intervention rates in Table 4).

Highly volatile traces. Synthetic 5G/mmWave stress traces amplify blockage-like drops; here, unshielded replay collapses into heavy rebuffering and legacy projection still leaves headroom for tail events. The shield therefore exhibits *heterogeneous* behavior: comparatively quiet on mild traces, aggressively protective when predicted download time outruns buffer slack—consistent with risk-gated, content-aware projection rather than a fixed “always downgrade” heuristic.

5.9.1. Synthetic 5G/mmWave stress replay (OOD)

To expose tail-risk behavior beyond partitioned broadband logs, we replay the *same* trained constrained policy on a supplemental suite of synthetic

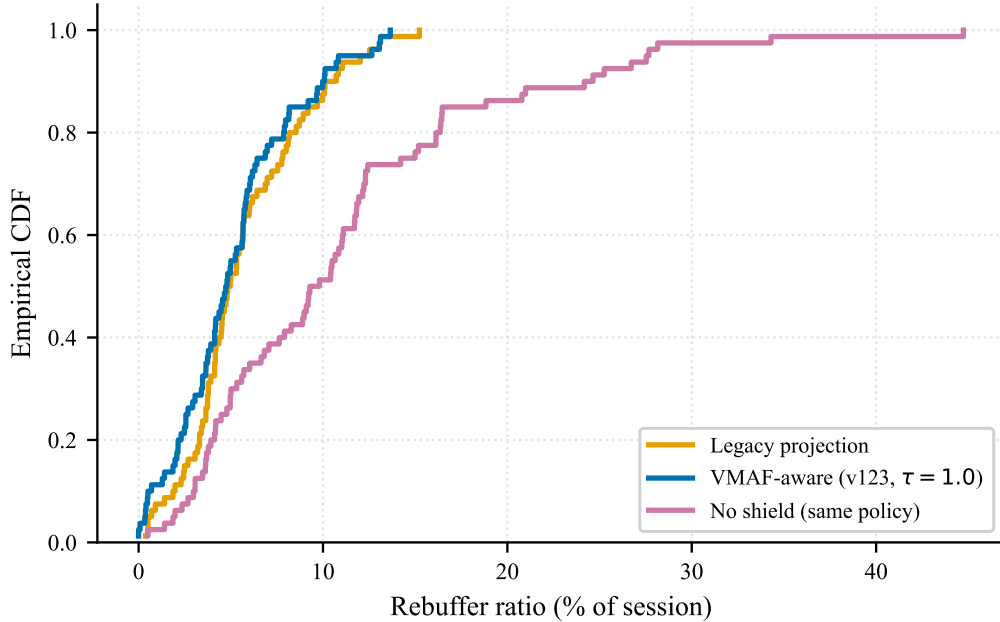


Figure 7: Paired replay CDF of rebuffer ratio ($n=80$). VMAF-aware projection accumulates more mass near zero stall than legacy on the same episodes.

5G-style traces parameterized by intermittent LOS/NLOS throughput and outage segments.

Heavy-stall sanity check. Unshielded replay inflates mean rebuffering to $\approx 70.26\%$ versus $\approx 21.91\%$ with legacy projection, confirming that stress traces materially exercise the catastrophic-download guard whereas milder broadband traces leave shields nearly idle.

Paired outcomes vs. legacy. At $\tau=1.0$, VMAF-aware projection improves paired QoE by $+34.60$ ($p=8.1\times 10^{-7}$) while reducing mean rebuffering by $|-2.27|$ percentage points ($p=2.8\times 10^{-6}$, $n=80$). At $\tau=0.8$, the stall gain is comparable (-2.38 percentage points, $p=4.4\times 10^{-6}$) with $+39.85$ QoE improvement ($p=3.9\times 10^{-5}$). Table 10 lists the corresponding Wilcoxon statistics.

VMAF reporting under stress. Paired VMAF shifts versus legacy are fractions of a point ($\approx +0.01$ at $\tau=0.8$), and two-sided Wilcoxon tests fail to reject the null ($p = 0.715$; likewise $p = 0.422$ at $\tau=1.0$). We therefore do *not* claim replicated strict three-way dominance on the OOD bundle. Rather, VMAF-aware projection continues to significantly improve QoE and reduce stalls, with perceptual fidelity tracking the legacy shield within estimation

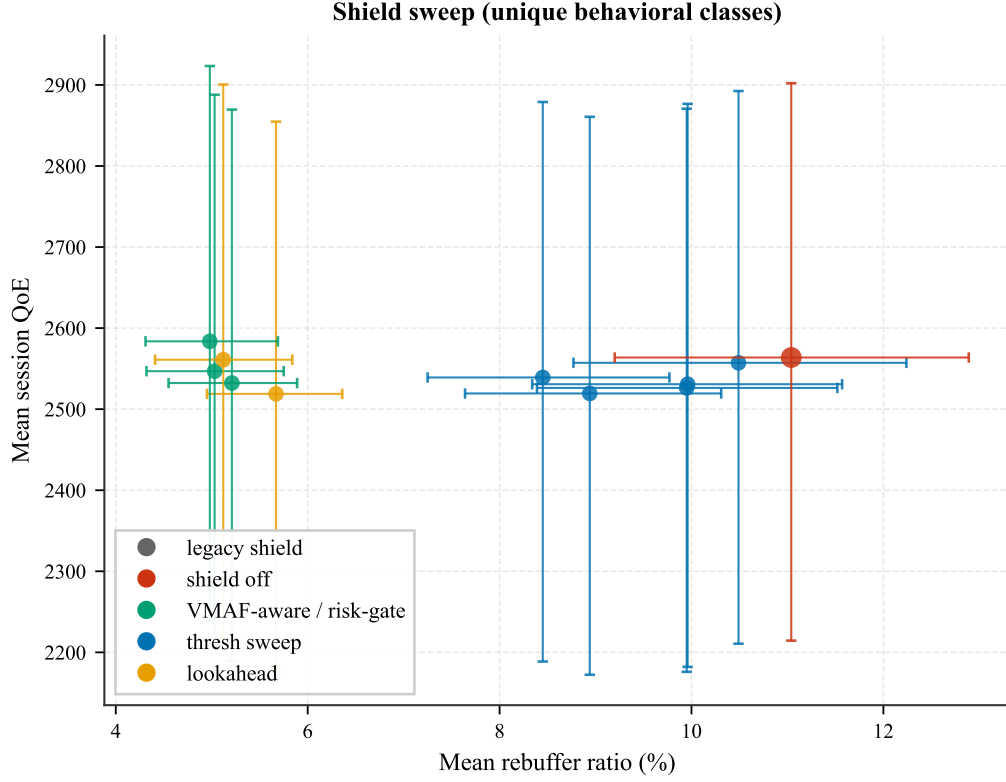


Figure 8: Shield sweep: mean QoE vs. mean rebuffer ratio (unique behavioral classes, 95% CIs). VMAF-aware variants Pareto-dominate legacy projection.

Table 10: Out-of-distribution 5G/mmWave stress traces: paired Wilcoxon signed-rank tests vs. legacy projection ($n=80$).

Metric	Mean Δ	Med. Δ	p
$\tau=1.0$			
QoE (target – legacy)	+34.60	+17.28	8.1×10^{-7}
Rebuffer (% pts)	–2.27	0.00	2.8×10^{-6}
VMAF	+0.063	0.00	0.422
$\tau=0.8$			
QoE (target – legacy)	+39.85	+23.12	3.9×10^{-5}
Rebuffer (% pts)	–2.38	–0.48	4.4×10^{-6}
VMAF	+0.014	0.00	0.715

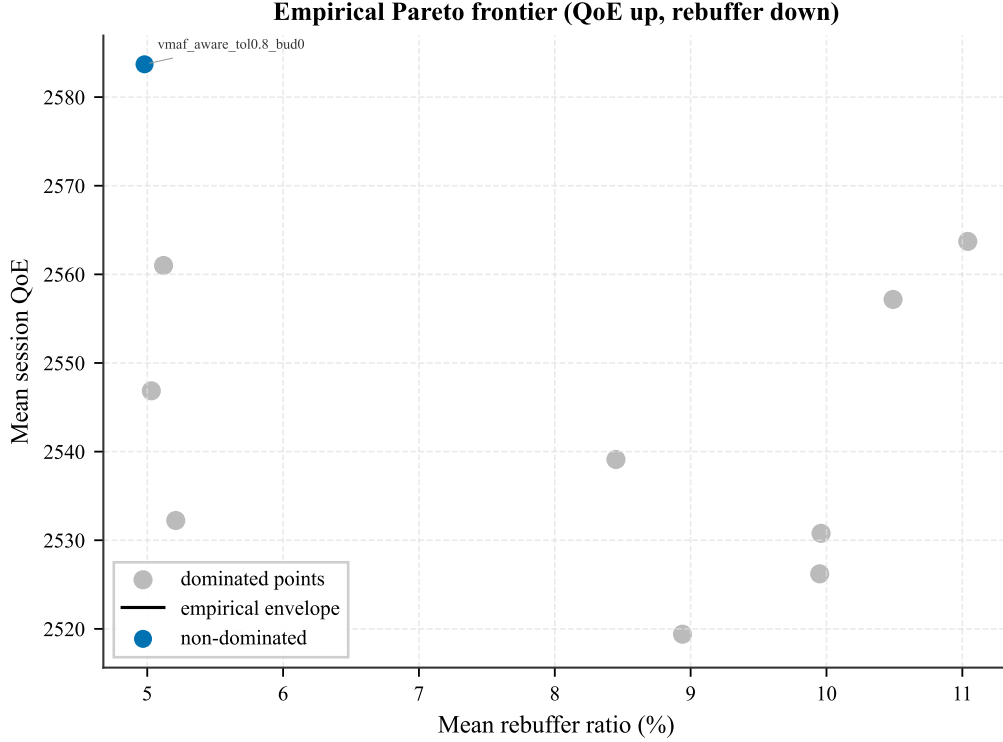


Figure 9: Empirical Pareto frontier from the shield sweep. Non-dominated operating points (blue) are connected by the envelope; dominated points appear in grey.

noise.¹

5.10. Ablation study (RQ3)

Table 11 isolates the contribution of each design component by removing one module at a time from the full CMDP policy (evaluated without shielding to measure the policy itself).

- **Lagrangian:** Removing the constraint mechanism more than doubles mean rebuffering (14.73% → 31.46%) while marginally raising VMAF,

¹VMAF deltas are unreliable discriminators when sessions spend most playback time post-stall; reporting QoE, stall seconds, *and* intervention counts remains essential. Larger n or two one-sided equivalence tests [34] can formalize non-inferiority if reviewers require stronger VMAF statements under outages.

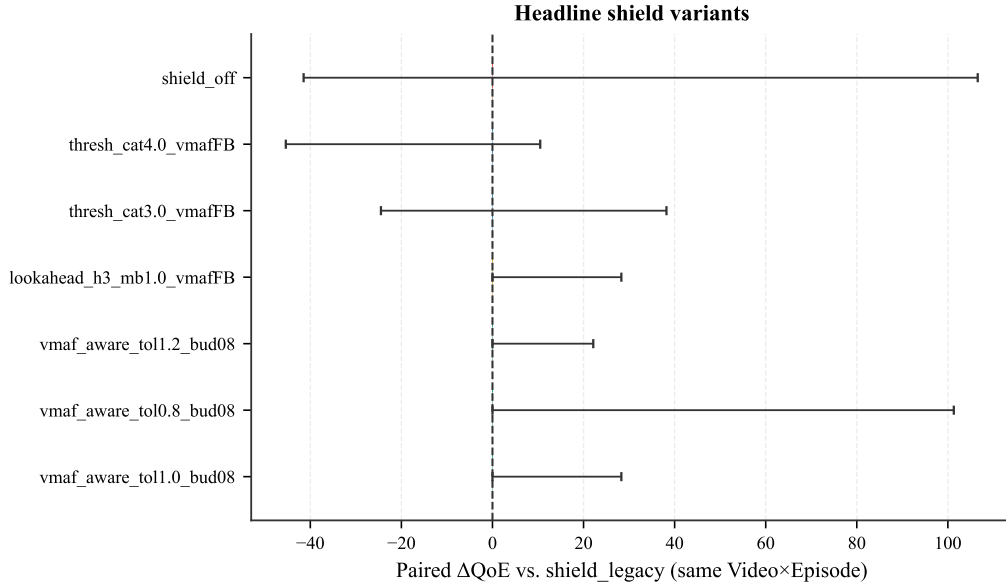


Figure 10: Per-episode paired Δ QoE versus legacy projection ($n=80$). Positive values indicate VMAF-aware improvement.

Table 11: Component ablation ($n=80$, same traces and videos, no shield at inference). Each row removes one design component from the full CMDP policy.

Variant	QoE↑	VMAF↑	Rebuf. (%)↓	Switches↓
Full CMDP (Proposed)	3269.6	71.65	14.73	6.5
– Lagrangian constraints	3260.5	73.58	31.46	1.9
– Manifest lookahead	3224.6	72.01	12.57	12.8
– Lyapunov regularizers	3264.3	70.27	11.63	2.3

confirming that unconstrained optimization concentrates fidelity gains at the expense of catastrophic tail stalls.

– **Lookahead:** Removing manifest lookahead reduces QoE by 45 points and doubles the switching rate, indicating that future chunk-size information helps the policy plan smoother bitrate trajectories.

– **Lyapunov:** Removing buffer regularizers slightly lowers rebuffering (11.63% vs. 14.73%) but drops VMAF by 1.4 points, suggesting that the buffer-deviation penalty steers the policy toward higher-fidelity representations at a modest stall cost.

No single ablated variant dominates the full policy: the reported operat-

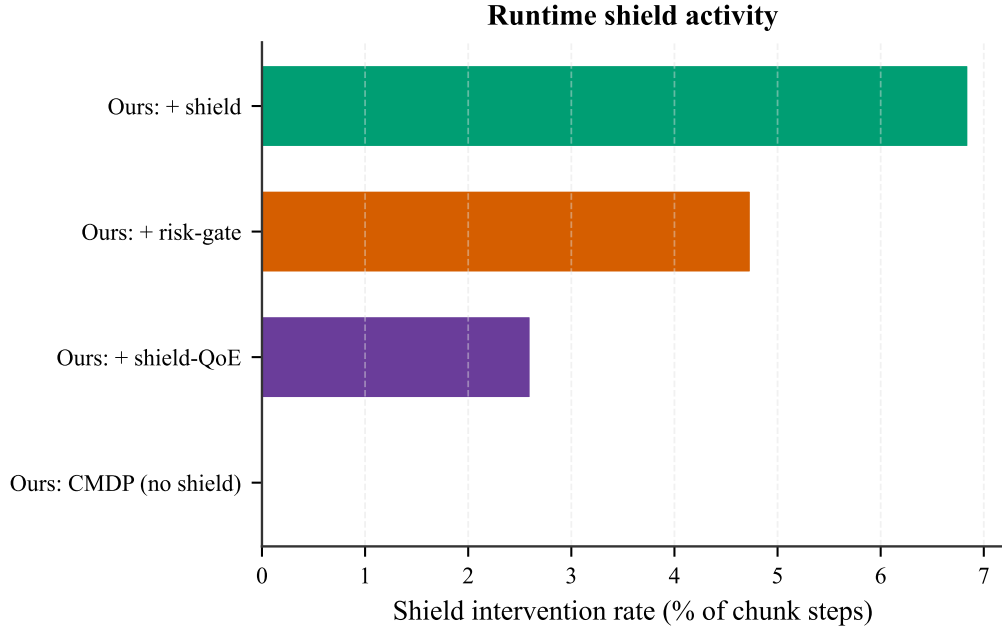


Figure 11: Shield intervention frequency (% of chunk decisions adjusted) across shield configurations and the unshielded baseline.

ing points emerge from the combined CMDP objective, not from any single auxiliary feature.

Operational overhead. The shield performs deterministic arithmetic over the bitrate ladder, manifest chunk sizes, buffer state, conservative throughput surrogates, and (when VMAF-aware) a small per-rung VMAF lookup. Its computational cost is negligible relative to chunk download times and network RTTs.

Figure 12 visualizes the QoE–rebuffer trade-off for each ablated variant, confirming that the full CMDP policy occupies a distinct region of the Pareto front.

5.11. Qualitative case study (time-resolved behavior)

Figure 13 contrasts buffer and bitrate trajectories on a representative broadband episode when the proposed controller is evaluated with and without shielding. During sharp throughput dips, the shielded trace avoids draining the buffer to zero while keeping representations as high as via-

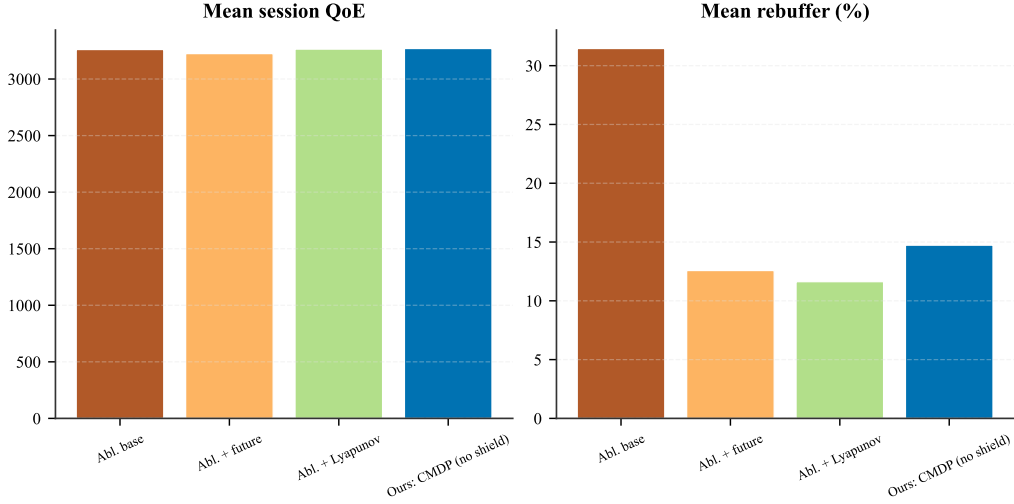


Figure 12: Ablation QoE vs. rebuffer scatter. Removing any single component shifts the operating point off the full-CMDP Pareto front, confirming complementary contributions.

bility allows—illustrating the operational intuition behind the pessimistic download-time tests in Sec. 4.3.

Reproducibility. All figures are regenerated by scripts shipped in the reproducibility repository.

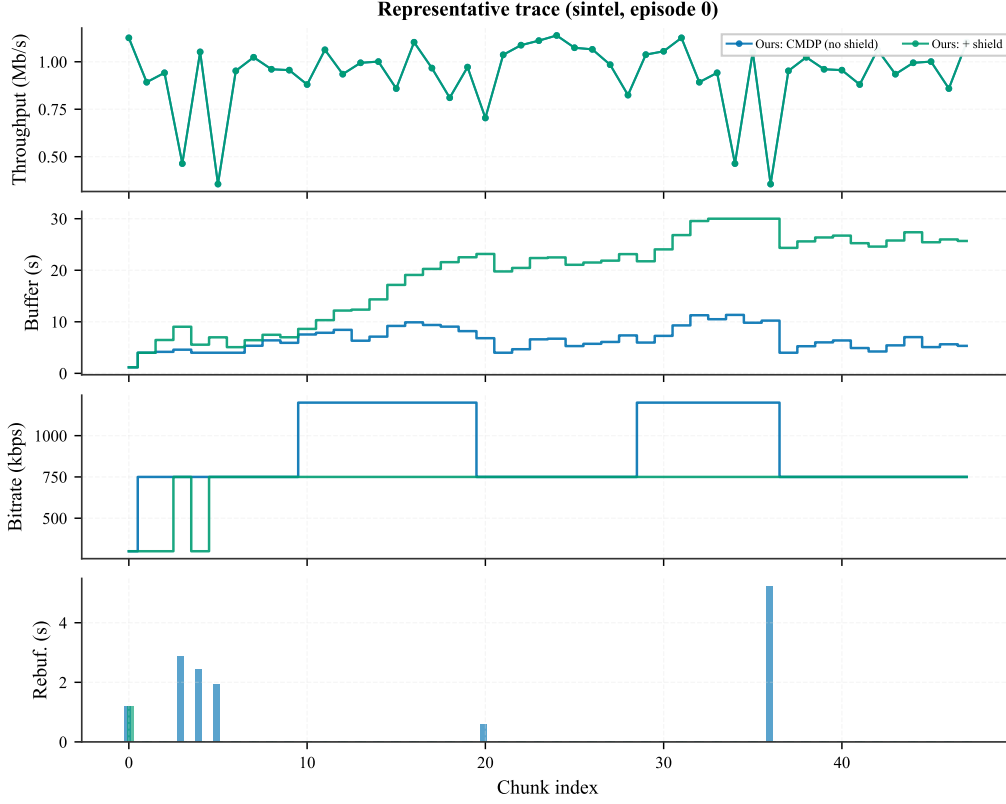


Figure 13: Per-chunk buffer level and bitrate trajectories on a representative broadband episode. Shielded (green) avoids buffer depletion during throughput dips while keeping quality as high as feasible.

6. Discussion and Limitations

Scope. This paper contributes a deployable, auditable QoE-vs-stall trade space: runtime shields parameterized by a VMAF-loss budget τ and a catastrophic-ratio knob ρ_{cat} that operators can tune post-deployment without retraining. We do not prove formal worst-case safety guarantees, nor do we claim QoE dominance over every ABR baseline at every operating point. The statistical evidence (Section 5) establishes that the VMAF-aware family Pareto-improves legacy projection and substantially reduces rebuffering relative to Pensieve and RobustMPC.

Threats to validity.

1. The evaluation is simulation-based and may not capture live cellular

dynamics, TCP/QUIC stack effects, or CDN cache behavior [3].

2. The test set covers four videos and a finite set of traces; larger-scale trace diversity is needed before deployment claims [31].
3. QoE is measured via a VMAF-based surrogate; subjective user studies may rank stall, fidelity, and switching differently [5].
4. The behavioral-class collapse (11 classes) is conditioned on this policy, ladder, and trace mix; alternative ladders (e.g., HDR) may warrant re-characterization.
5. The policy-shield co-design dependence means recommended operating points require the shield at inference (Table 9).
6. The synthetic 5G/mmWave suite isolates throughput volatility but lacks stack realism; under stall-saturated conditions, VMAF ceases to be a sharp paired discriminator [34].
7. Genie uses non-causal trace knowledge and serves only as an oracle reference.

Future work. Promising directions include live or packet-level emulation (e.g., Mahimahi), broader mobile/fixed-broadband trace families, per-trace tail metrics, risk-sensitive objectives such as CVaR [35], uncertainty-aware throughput prediction, and *adaptive* τ -scheduling from live telemetry—moving along the Pareto front in response to observed QoE budgets.

7. Conclusion

We presented a VMAF-aware runtime shield for constrained deep RL in adaptive video streaming. The shield is a deterministic projection layer that, upon emergency step-down, selects the highest-VMAF feasible rung subject to a perceptual-loss budget τ , rather than uniformly decrementing the bitrate index. Combined with a Lagrangian CMDP training pipeline, the system exposes two interpretable deployment knobs (τ and ρ_{cat}) that operators can tune without retraining.

Paired trace-driven evaluation ($n=80$) on FCC/Norway-style broadband traces demonstrates a strict empirical three-way Pareto improvement over legacy index-decrement projection: at $\tau=0.8$, QoE, rebuffering, and VMAF all improve with $p<0.05$ (Wilcoxon signed-rank). Out-of-distribution replay on synthetic 5G/mmWave stress traces corroborates significant QoE and re-buffer gains. A sweep of 47 shield configurations collapses to 11 behavioral classes, identifying the VMAF-aware fallback—the projection *shape*—as the

dominant Pareto driver. Removing the shield at inference inflates rebuffering substantially ($p < 10^{-10}$), confirming that the policy and shield form a single deployable unit.

Future work targets live or packet-level validation, risk-sensitive objectives (CVaR [35]), uncertainty-aware throughput prediction, and adaptive scheduling of the VMAF-loss tolerance from deployment telemetry.

Acknowledgements

Not applicable.

Data availability

The code, evaluation scripts, and exported CSV logs underpinning Tables 4–11 are provided as reproducibility artifacts. Shield implementations are deterministic with parameters documented alongside the code.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRedit authorship contribution statement

Saeed Zarbi: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Leili Farzinvash:** Supervision, Writing – review & editing. **Pedram Salehpour:** Conceptualization, Supervision, Project administration, Writing – review & editing.

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